1. Purpose

- This SOP provides ORL's standard guidelines and procedure for Inspection of all incoming material at site. Material inspection process involves checking of various aspects of material like physical and chemical properties to confirm that the received material is as per the required specifications of approved material as mentioned in Material approval submittal (MAS) for contractor procured material and as per HO standard approval for ORL procured material.

2. Scope of work

- This Standard operating procedure addresses the process of receiving, sampling, inspection, testing, and storage of the incoming materials at site and approval along with the responsibility matrix.

3. Responsibility Matrix

3.1 For Contractor's Material

Activities	Responsible	Accountable	Consulted	Informed
Receiving of the materials at site	Contractor	Store In-charge	Project QC / Execution In-charge	Project team
Sampling and Inspection of the materials.	Contractor	Project QC personnel	Project QC / Execution In-charge	Project team
Testing as per approved ITP	Contractor	Project QC personnel	Project QC / Execution In-charge	Project team
Proper storage of the materials	Contractor	Store In-charge	Project QC / Execution In-charge	Project team

NOTE*	
Responsible	The person who performs an activity or does the work.
Accountable	The person who is ultimately accountable and has YES/NO/VETO.
Consulted	The person that needs to give feedback and contribute to the action.
Informed	The person that needs to know of the decision/action.

3.2 For ORL Material

Activities	Responsible	Accountable	Consulted	Informed
Receiving of the materials at site.	Store In-charge	Site Execution team	Project QC team	Project team
Sampling and Inspection of the materials.	Site Execution Team	Project QC personnel	Project QC/ Execution In-charge	Project team
Testing as per ITP	Site Execution Team	Project QC personnel	Project QC/ Execution In-charge	Project team
Proper storage of the raw materials	Store In-charge	Store In-charge	Project QC/ Execution team	Project team

NOTE*	
Responsible	The person who performs an activity or does the work.
Accountable	The person who is ultimately accountable and has YES/NO/VETO.
Consulted	The person that needs to give feedback and contribute to the action.

Informed	The person that needs to know of the decision/action.

4. Method of work

4.1. Contractor's Incoming Material Inspection

- Once the material delivered at site the security personnel shall inform the contractor's store In-charge.
- The Contractor's store in-charge will perform the material check for 'Quantity' and raise 'Material Inspection Request (MIR)' through digital system.
- The contractors store In-charge shall inform to contractor's Quality team/ Site person/MEP Person.
- Contractor's Site QC Engineer to train Site SPOC Engg. (Civil/ MEP) for material inspection.
- The contractor's QC Engineer/ (SPOC Engg. – Civil/ MEP) shall inspect the material as per the approved ITP/MAS.
- After internal inspection contractor to assign material inspection request to ORL project QC Engineer / (SPOC Engg. – Civil/ MEP) through digital system.
- MIR shall include following documents for inspection:
  - MIR form
  - Delivery challan, lot no., batch no.
  - Manufacturer test certificate/ Internal lab test report/third party test report.
  - Material approval copy/reference number.
  - ITP reference number -to conform test are conducted as per requirement.
  - Factory acceptance test / report, if applicable
  - Picture indicating Lot number /stacking of material.
  - Country of origin certificate, if applicable
- ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) to confirm the date and time for material inspection on digital system.
- ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) shall physically inspect the material as per the approved ITP.
- ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) shall verify the MTC along with the challans for all semi-processed / manufactured incoming material.
- ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) shall verify that the incoming material is as per material approved submittal. Without MTC's, no material shall be accepted. MTC shall be match with the lot number of the material.
- Field test shall be carried out for material as per approved ITP and data shall be recorded with pictures in MIR.
- For the materials whose MTC are not provided at the time of delivery of material, the Project ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) shall accept the material on the basis of previous Third-Party testing done as per ITP and Contractor shall submit its MTC within 07 days from the receipt of material.
- After approval of MIR from ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP), contractor shall unload the material at site.
- ORL Project QC Engineer / (SPOC Engg. – Civil/ MEP) shall reject the material which do not comply with the required specifications.
- Rejected material shall be segregated and stored separately with proper rejection tagging till it is removed from the site with proper out gate pass signed by ORL representative.
- The materials shall be sent for third party testing as per defined frequencies in the approved Inspection and Test Plans (ITP).
- Third party testing shall be done at NABL/ORL approved laboratory only.

- Sample chosen for testing shall be representative and shall be selected in confirming to the relevant Indian/ International standards. Sampling shall be done in presence of ORL QC Engineer / (SPOC Engg. – Civil/ MEP) and signoff to be taken.
- In case the material doesn't meet the specifications in the third-party NABL lab then the unloaded material shall keep on hold and discarded from site.
- The materials should be stored properly as per the process mentioned in the method statement approved by ORL project QC Engineer / (SPOC Engg. – Civil/ MEP) and material shall be consumed first in first out and shall be consumed within its shelf life.
- Procedures for the handling, packing, marking, preservation, storage, and delivery of material shall be in place for all materials by considering the prevailing weather conditions on the construction site and the likely conditions encountered during transit.
- The storage guidelines for various construction materials shall be as per Manufacturer recommendations and IS 4082.
- Pre dispatch factory Inspections can also be done at suppliers' premises and the same shall be witnessed by ORL / contractor representative, as appropriate.

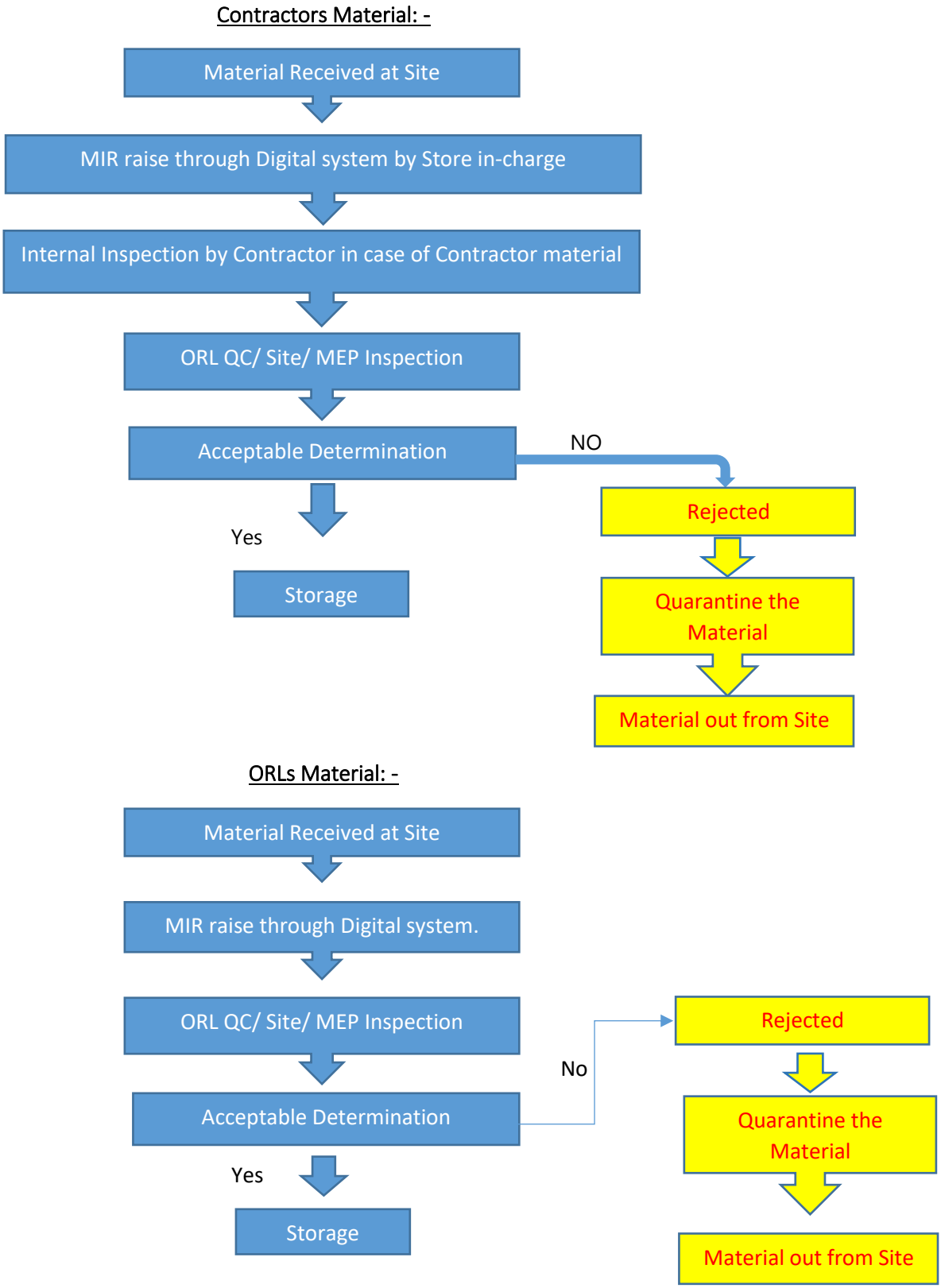
4.2. ORL Material

- Once the material delivered at site the security personnel shall inform ORL store In-charge.
- The ORL's store in-charge will perform the material check for 'Quantity' and raise 'Material Inspection Request (MIR)' through digital system.
- The Store In-charge shall inform ORL project QC Engineer / (SPOC Engg. – Civil/ MEP).
- The ORL project QC Engineer / (SPOC Engg. – Civil/ MEP) shall inspect the material as per the HO approval and approved ITP.
- MIR shall include following documents for inspection:
  - MIR form
  - Delivery challan, lot no., batch no.
  - Manufacturer test certificate/ Internal lab test report/third party test report.
  - Material approval copy/reference number.
  - ITP reference number -to conform test are conducted as per requirement.
  - Factory acceptance test / report, if applicable
  - Picture indicating Lot number /stacking of material.
  - Country of origin certificate, if applicable
- ORL project QC Engineer / (SPOC Engg. – Civil/ MEP) shall physically inspect the material as per the approved ITP.
- ORL project QC Engineer / (SPOC Engg. – Civil/ MEP)l shall verify the MTC along with the challans for all semi-processed / manufactured incoming material.
- ORL project QC Engineer / (SPOC Engg. – Civil/ MEP)l shall verify that the incoming material is as per HO approval. Without MTC's, no material shall be accepted. MTC shall be match with the lot number of the material.
- Field test shall be carried out for material as per approved ITP and data shall be recorded with pictures in MIR.
- For the materials whose MTC are not provided at the time of delivery of material, the Project ORL QC Engineer / (SPOC Engg. – Civil/ MEP) shall accept the material on the basis of previous Third-Party testing done as per ITP and Vendor/ Supplier shall submit its MTC within 07 days from the receipt of material.

- After approval of MIR from ORL Quality personnel, ORL store personal shall unload the material at site.
- ORL QC Engineer / (SPOC Engg. – Civil/ MEP) shall reject the material which do not comply with the required specifications.
- Rejected material shall be segregated and stored separately with proper rejection tagging till it is removed from the site with proper out gate pass signed by ORL representative.
- The materials shall be sent for third party testing as per defined frequencies in the approved Inspection and Test Plans (ITP).
- Third party testing shall be done at NABL/ORL approved laboratory only.
- Sample chosen for testing shall be representative and shall be selected in confirming to the relevant Indian/ International standards. Sampling shall be done in presence of ORL Quality Engineer and signoff to be taken.
- In case the material doesn't meet the specifications in the third-party NABL lab then the unloaded material shall keep on hold and discarded from the site.
- The materials should be stored properly as per the process mentioned in the method statement approved by ORL QC Engineer / (SPOC Engg. – Civil/ MEP) and material shall be consumed first in first out and shall be consumed within its shelf life.
- Procedures for the handling, packing, marking, preservation, storage, and delivery of material shall be in place for all materials by considering the prevailing weather conditions on the construction site and the likely conditions encountered during transit.
- The storage guidelines for various construction materials shall be as per Manufacturer recommendations and IS 4082.
- Pre dispatch factory Inspections can also be done at suppliers' premises and the same shall be witnessed by ORL representative, as appropriate.

5.0 Flow chart:

A flow chart indicating the process of workflow is mentioned below.



6.0 Frequency

The above procedures shall be followed during inspection of all incoming materials delivered to site.

7.0 Reference Documents:

A. ITP- Inspection test plan
B. Contact Specifications
C. Material Approval Submittal (MAS)
D. HO approval
E. Project Consultant Specifications
F. IS: 4082 – Stacking and Storage of Construction Materials and Components at site - Recommendations